

JINU ROSE MATHEW

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SKILLS

Programming & Analytics	Python (Pandas, NumPy, Scikit-learn), SQL
Business Intelligence & GIS	Power BI, IBM Cognos Analytics, Tableau, ArcGIS Pro
Databases & Cloud	Oracle, MySQL, PostgreSQL, AWS (Glue, Redshift)
Tools	Git, Jupyter Notebook

EXPERIENCE

Research Analyst Austin 311, City of Austin	Dec 2024 - Present <i>Austin, TX</i>
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- Performed daily system monitoring and data validation for the CSR (Customer Service Request) platform supporting city-wide 311 operations, ensuring high system availability and data integrity.
- Led monthly spatial data enhancement and validation processes using ArcGIS Pro, including spatial joins, attribute calculations, and quality checks across Address Point and Transportation Street Segment layers.
- Managed geospatial data updates by reviewing modified layers, applying transformations, and exporting validated datasets to enterprise geodatabases.
- Designed and delivered Power BI dashboards and IBM Cognos reports, including a Capital Improvement Project prioritization model to support data-driven infrastructure planning.
- Configured and maintained Service Request (SR) workflows, implementing minor system enhancements and configuration changes to align with evolving business requirements.
- Created and maintained detailed work instructions and standard operating procedures (SOPs) to improve process standardization, knowledge transfer, and audit readiness.

Program Manager Additional Skill Acquisition Program	Jan 2017 - Sep 2019 <i>Thiruvananthapuram, Kerala, India</i>
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- Led stakeholder requirement gathering and cross-functional delivery of MIS application updates.
- Automated weekly and executive reports using Python and SQL Server, reducing manual reporting effort.
- Managed release cycles including integration, functional, UAT, and smoke testing.
- Developed technical documentation in Confluence to support system maintenance and onboarding.

PROJECTS

- **Retrieval Augmented Generation based Restaurant Chatbot with AI Testability:** Developed an advanced Restaurant Chatbot employing Retrieval Augmented Generation. Built knowledge graph using Neo4j for response retrieval and applied NLP using T-5 large language model. Ensured robust and accurate responses with intelligent testing capabilities and implemented an intuitive user interface to enhance the user interaction.
- **Analysis on Side Effects of COVID-19 Vaccination:** Designed a data model using MySQL Workbench to analyze the adverse side effects of various vaccines for Covid-19. Carried out data preprocessing using Pandas and analyzed the dataset in PostgreSQL using SQL queries and Tableau.
- **Exploratory Data Analysis of NYC TLC:** Created and executed a data pipeline using Python and AWS Glue for analyzing the NYC trip taxi records. Developed a dashboard to analyze the dataset using graphical representation by connecting Redshift with Tableau. Retrieval Augmented Generation based Restaurant Chatbot with AI Testability

EDUCATION

Master of Science in Data Analytics , San Jose State University	Dec 2023
Core Areas - Data Visualization, Data Mining, Machine Learning and Big Data	GPA: 3.88

Bachelor of Technology in Computer Science , Mahatma Gandhi University	May 2015
Amal Jyothi College of Engineering	